

Business request health insurance analytics · MD

Business Request: Health Insurance Cost & Claims Analytics

To: Data Analyst / Data Scientist Team **From:** VP of Actuarial & Business Analytics
Date: August 2026 **Priority:** High **Project Code:** HIC-2026-001

1. Background

Our health insurance division has accumulated several years of policyholder-level claims data, including demographic details (age, gender), health risk indicators (BMI, blood pressure, diabetic status, smoker status), family structure (number of children), regional distribution, and the resulting claim amounts.

Currently, this data is reviewed manually on an ad-hoc basis, with no centralized, self-service reporting tool. Regional managers, underwriting teams, and executive leadership each request different cuts of this data independently, creating duplicated effort and inconsistent figures across departments. We currently lack the ability to proactively identify high-risk policyholder segments or forecast future claims exposure.

2. Problem Statement

The business does not have real-time visibility into:

- Which policyholder segments (by region, smoking status, BMI category, or age group) are driving the highest claims costs
- How claims costs are trending across demographic and risk factors
- Which policyholders are likely to file high-cost claims in the future, before those costs are incurred
- A consistent, governed source of truth for claims reporting that non-technical stakeholders can use independently

This lack of visibility limits our ability to price policies accurately, allocate reserves appropriately, and design targeted wellness or risk-mitigation programs.

3. Business Objective

Leverage our existing claims dataset to:

1. Build a self-service **Power BI dashboard** that gives underwriting, regional managers, and executive leadership immediate visibility into claims performance by demographic and risk segment.
2. Apply **data science / predictive modeling** to forecast expected claim costs and flag high-risk policyholders, using domain knowledge of health insurance risk factors (smoking, BMI, age, pre-existing conditions such as diabetes).
3. Package the predictive model into an accessible **application/tool** so non-technical business users (underwriters, account managers) can input policyholder attributes and receive an estimated claim risk score — without needing to understand the underlying model.

4. Scope of Work

Phase 1 — Data Preparation & Dashboard (Power BI)

- Clean and validate the claims dataset (handle missing values, standardize categorical fields, verify data types)
- Design and build a multi-page Power BI dashboard covering:
 - Executive summary (KPIs: total claims, average claim, policyholder count, % high-risk segments)
 - Claims drivers by region, smoker status, diabetic status, and BMI category
 - Demographic breakdown (age, gender, family size)
- Deploy dashboard to Power BI Service with role-based access for regional managers vs. executive leadership

Phase 2 — Predictive Modeling

- Using domain knowledge of insurance risk factors, build a model to predict expected claim amount (or claim risk tier: low/medium/high) based on age, BMI, blood pressure, smoker status, diabetic status, children, and region
- Validate model performance against historical claims data
- Document key risk drivers identified by the model in business-friendly terms (not just statistical output)

Phase 3 — Prediction Application

- Build a lightweight internal tool/app where a user can enter a prospective or existing policyholder's attributes (age, BMI, smoker status, etc.) and receive:
 - An estimated claim amount or risk tier
 - A plain-language explanation of the key factors driving that prediction
- Integrate or link this tool with the Power BI dashboard where feasible, so business users have one consistent entry point

5. Deliverables

Deliverable	Description
Cleaned dataset	Validated, analysis-ready claims data
Power BI dashboard	Multi-page report with KPIs, trend analysis, and segment breakdowns
Predictive model	Trained and validated model estimating claim cost/risk
Prediction tool/app	User-facing interface for entering policyholder attributes and viewing predictions
Documentation	Summary of methodology, key findings, and risk drivers in non-technical language

6. Success Criteria

- Dashboard adopted as the single source of truth for claims reporting across underwriting and regional teams
- Predictive model identifies high-risk segments with sufficient accuracy to support underwriting decisions
- Business users (non-technical) can independently use the prediction tool without analyst support
- Reduction in time spent producing ad-hoc claims reports for leadership

7. Required Skill Set

The assigned analyst/data scientist should have:

- Strong proficiency in Python (pandas, data cleaning, statistical/ML modeling)
- Power BI dashboard design and DAX proficiency
- Domain knowledge of health insurance risk factors and claims behavior
- Ability to translate technical model output into business-relevant insights
- Experience building lightweight applications or interfaces for non-technical end users (a plus)

8. Timeline

Phase	Estimated Duration
Data cleaning & dashboard build	2 weeks
Predictive model development & validation	2–3 weeks
Prediction app/tool build	1–2 weeks
Stakeholder review & deployment	1 week

Requested by: Valuelink, VP of Actuarial & Business Analytics **Assigned to:** Emeka.V

